Figure 1. Banner

Centralized Security Monitoring System (SIEM Implementation using Wazuh)

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1. Project Title

Implementation of a Centralized Security Monitoring Solution Using Wazuh

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2. Objective

To implement a centralized security monitoring solution that collects, analyzes, and generates alerts from logs across multiple systems in real time, enhancing the organization's ability to detect, investigate, and respond to cyber threats effectively.

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3. Project Overview

This project focuses on deploying a Security Information and Event Management (SIEM) system using Wazuh on a dedicated infrastructure.

The solution enables:

- Centralized log collection
- Real-time threat detection
- Security event correlation
- Incident response support

The system is deployed on a dedicated HP PC running Ubuntu Server (TTY-only environment) to ensure stability, performance, and reduced attack surface.

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4. System Architecture

4.1 Components

- Wazuh Manager (Central server)
- Wazuh Agents (Installed on monitored endpoints)
- Elastic Stack / Wazuh Dashboard
- Ubuntu Server (TTY-only)

4.2 Architecture Diagram

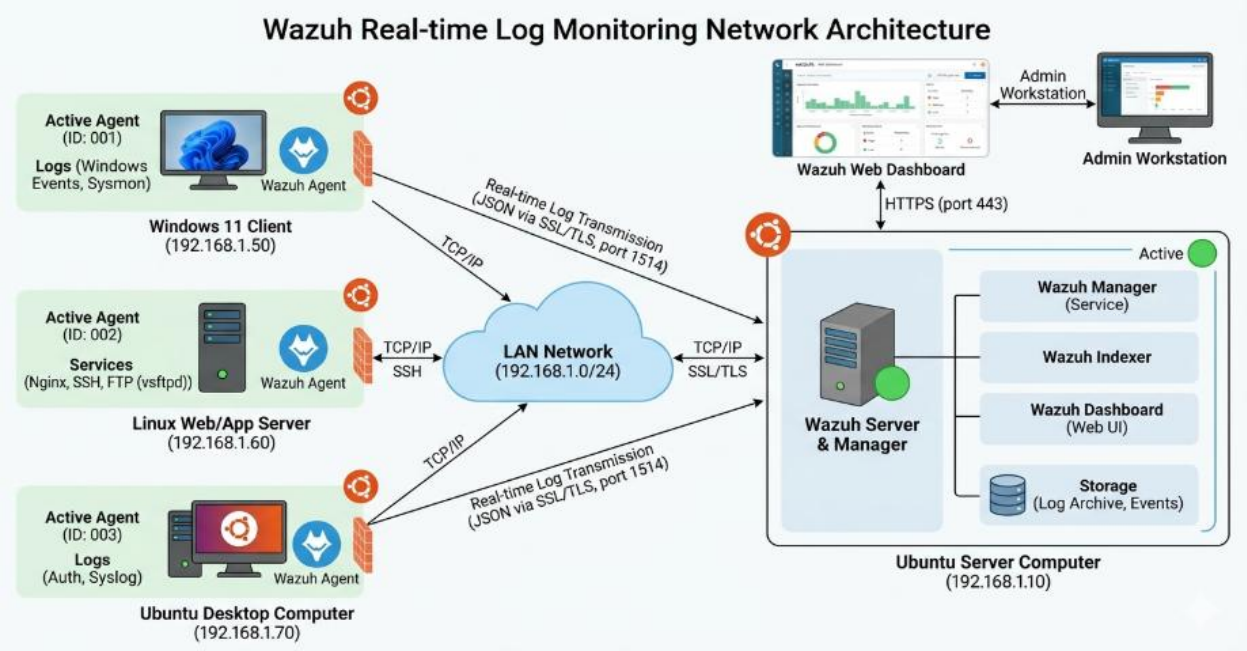

Figure 2. Deployment Architecture

5. Hardware & Software Requirements

5.1 Hardware

- Dedicated HP PC
- Minimum Requirements:
    - 4 GB RAM (8 GB recommended)
    - 100 GB Storage
    - Stable network connection

5.2 Software

- Ubuntu Server (TTY-only installation)
- Wazuh Manager
- Wazuh Agents
- Wazuh Dashboard

6. Installation & Setup

6.1 Ubuntu Server Installation (TTY Mode)

- Installed Ubuntu Server without GUI
- Configured:
  - Static IP address
  - SSH access
  - System updates

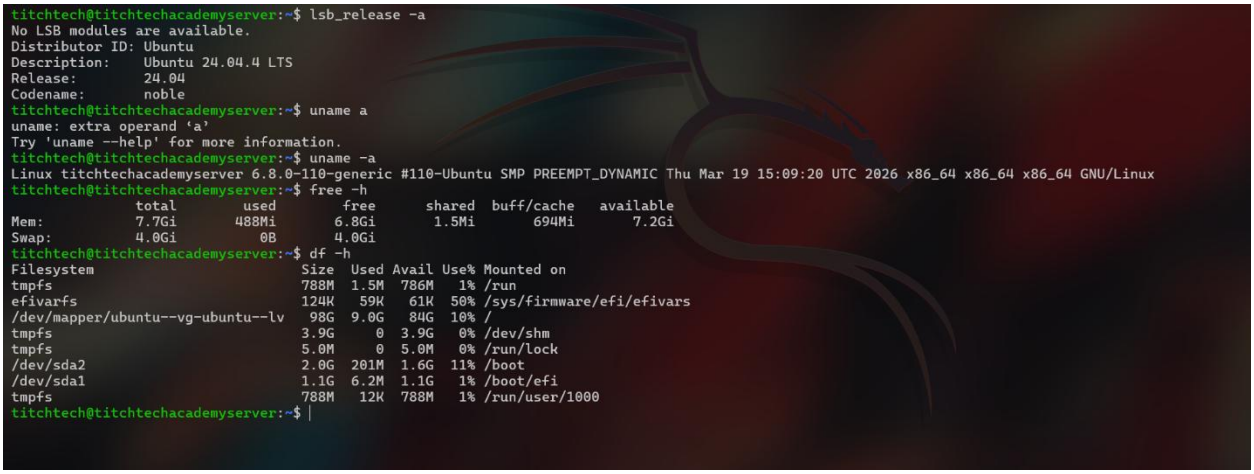

Figure 3. Server Sperks

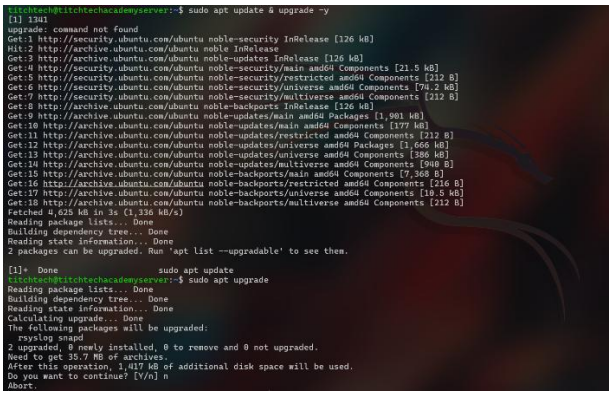

Figure 5. Update Server

Figure 4. Enable ssh service.

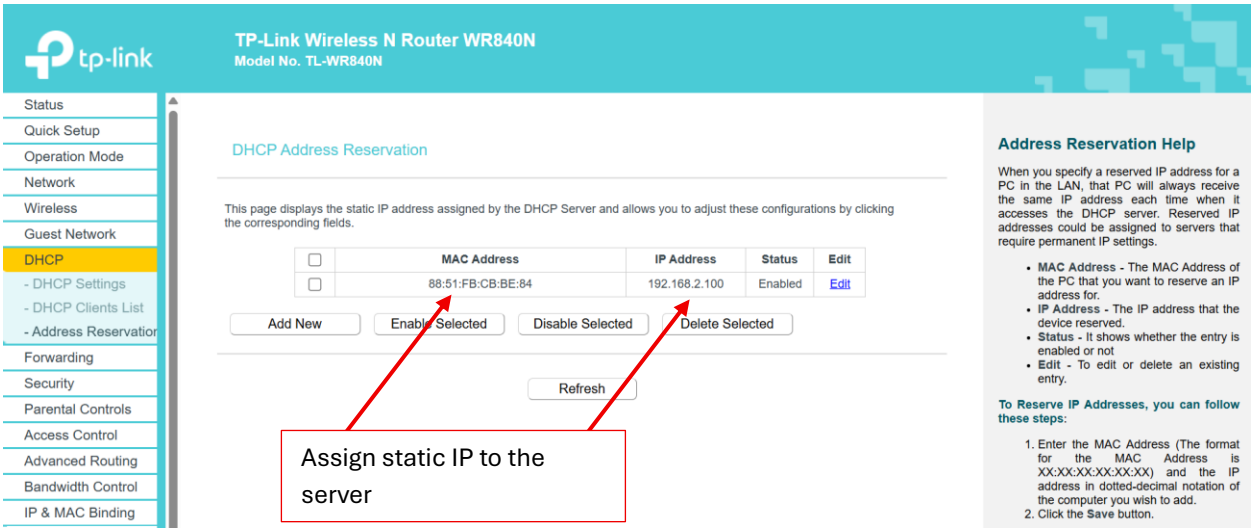

Figure 6. Server Static IP Adress

6.2 Wazuh Server Installation

- Installed Wazuh Manager on Ubuntu Server
- Verified services:
  - wazuh-manager
  - wazuh-api
  - wazuh-indexer
  - wazuh-dashboard

```
titchtech@titchtechacademyserver:~$ sudo bash wazuh-install.sh -a --ignore-check --overwrite
19/04/2026 03:50:38 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
19/04/2026 03:50:38 INFO: Verbose logging redirected to /var/log/wazuh-install.log
19/04/2026 03:50:39 INFO: --- Removing existing Wazuh installation ---
19/04/2026 03:50:39 INFO: Wazuh GPG key not found in the system
19/04/2026 03:50:40 INFO: Installation cleaned.
19/04/2026 03:50:46 WARNING: Hardware and system checks ignored.
19/04/2026 03:50:46 INFO: Wazuh web interface port will be 443.
19/04/2026 03:50:54 INFO: Wazuh repository added.
19/04/2026 03:50:54 INFO: --- Configuration files ---
19/04/2026 03:50:54 INFO: Generating configuration files.
19/04/2026 03:50:56 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
19/04/2026 03:50:56 INFO: --- Wazuh indexer ---
19/04/2026 03:50:56 INFO: Starting Wazuh indexer installation.
19/04/2026 03:52:42 INFO: Wazuh indexer installation finished.
19/04/2026 03:52:42 INFO: Wazuh indexer post-install configuration finished.
19/04/2026 03:52:42 INFO: Starting service wazuh-indexer.
19/04/2026 03:53:00 INFO: wazuh-indexer service started.
19/04/2026 03:53:00 INFO: Initializing Wazuh indexer cluster security settings.
19/04/2026 03:53:11 INFO: Wazuh indexer cluster initialized.
19/04/2026 03:53:11 INFO: --- Wazuh server ---
19/04/2026 03:53:11 INFO: Starting the Wazuh manager installation.
19/04/2026 03:54:27 INFO: Wazuh manager installation finished.
19/04/2026 03:54:27 INFO: Starting service wazuh-manager.
19/04/2026 03:54:46 INFO: wazuh-manager service started.
19/04/2026 03:54:46 INFO: Starting Filebeat installation.
19/04/2026 03:55:26 INFO: Filebeat installation finished.
19/04/2026 03:55:34 INFO: Filebeat post-install configuration finished.
19/04/2026 03:55:34 INFO: Starting service filebeat.
19/04/2026 03:55:35 INFO: filebeat service started.
19/04/2026 03:55:35 INFO: --- Wazuh dashboard ---
19/04/2026 03:55:35 INFO: Starting Wazuh dashboard installation.
19/04/2026 03:57:25 INFO: Wazuh dashboard installation finished.
19/04/2026 03:57:25 INFO: Wazuh dashboard post-install configuration finished.
19/04/2026 03:57:25 INFO: Starting service wazuh-dashboard.
19/04/2026 03:57:26 INFO: wazuh-dashboard service started.
19/04/2026 03:58:00 INFO: Initializing Wazuh dashboard web application.
19/04/2026 03:58:02 INFO: Wazuh dashboard web application initialized.
19/04/2026 03:58:02 INFO: --- Summary ---
19/04/2026 03:58:02 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password: IKfNZ4KKArG44H818jH.Xugw4QYX*8PW
19/04/2026 03:58:02 INFO: Installation finished.
titchtech@titchtechacademyserver:~$
```

Figure 7. Wazuh Installation

```
titchtech@titchtechacademyserver:~$ sudo systemctl status wazuh-dashboard.service
● wazuh-dashboard.service - wazuh-dashboard
     Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-04-19 03:57:43 UTC; 3min 20s ago
   Main PID: 49604 (node)
      Tasks: 11 (limit: 9346)
     Memory: 164.4M (peak: 258.2M)
        CPU: 11.909s
     CGroup: /system.slice/wazuh-dashboard.service
             └─49604 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-rejections=warn /usr/share/wazuh-dashboard/src/cli/d

Apr 19 03:57:51 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:51Z","tags":["info","savedobjects-service"],"pid":49604
Apr 19 03:57:51 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:51Z","tags":["info","savedobjects-service"],"pid":49604
Apr 19 03:57:53 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:53Z","tags":["info","savedobjects-service"],"pid":49604
Apr 19 03:57:53 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:53Z","tags":["info","savedobjects-service"],"pid":49604
Apr 19 03:57:53 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:53Z","tags":["info","plugins-system"],"pid":49604,"mess
Apr 19 03:57:54 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:54Z","tags":["error","opensearch","data"],"pid":49604,"
Apr 19 03:57:54 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:54Z","tags":["error","opensearch","data"],"pid":49604,"
Apr 19 03:57:54 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:54Z","tags":["listening","info"],"pid":49604,"message":
Apr 19 03:57:55 titchtechacademyserver opensearch-dashboards[49604]: {"type":"log","@timestamp":"2026-04-19T03:57:55Z","tags":["info","http","server","OpenSearchDashboar
Apr 19 03:58:02 titchtechacademyserver opensearch-dashboards[49604]: {"type":"response","@timestamp":"2026-04-19T03:58:00Z","tags":[],"pid":49604,"method":"get","statusC
lines 1-20/20 (END)
```

Figure 9. Check Dashboard Servie Status

```
titchtech@titchtechacademyserver:~$ sudo systemctl status wazuh-manager.service
● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-04-19 03:54:46 UTC; 5min ago
      Tasks: 141 (limit: 9346)
     Memory: 719.0M (peak: 720.0M)
        CPU: 47.668s
     CGroup: /system.slice/wazuh-manager.service
             ├─7361 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
             ├─47400 /var/ossec/bin/wazuh-authd
             ├─47416 /var/ossec/bin/wazuh-db
             ├─47428 /var/ossec/bin/wazuh-execd
             ├─47442 /var/ossec/bin/wazuh-analysisd
             ├─47455 /var/ossec/bin/wazuh-syscheckd
             ├─47471 /var/ossec/bin/wazuh-remoted
             ├─47488 /var/ossec/bin/wazuh-logcollector
             ├─47499 /var/ossec/bin/wazuh-monitord
             ├─47511 /var/ossec/bin/wazuh-modulesd
             ├─48158 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
             ├─48161 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
             └─48164 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py

Apr 19 03:54:38 titchtechacademyserver env[47305]: Started wazuh-db...
Apr 19 03:54:39 titchtechacademyserver env[47305]: Started wazuh-execd...
Apr 19 03:54:40 titchtechacademyserver env[47305]: Started wazuh-analysisd...
Apr 19 03:54:41 titchtechacademyserver env[47305]: Started wazuh-syscheckd...
Apr 19 03:54:41 titchtechacademyserver env[47305]: Started wazuh-remoted...
Apr 19 03:54:42 titchtechacademyserver env[47305]: Started wazuh-logcollector...
Apr 19 03:54:43 titchtechacademyserver env[47305]: Started wazuh-monitord...
Apr 19 03:54:44 titchtechacademyserver env[47305]: Started wazuh-modulesd...
Apr 19 03:54:46 titchtechacademyserver env[47305]: Completed.
Apr 19 03:54:46 titchtechacademyserver systemd[1]: Started wazuh-manager.service - Wazuh manager.
```

Figure 10. Check Manager Service Status

```
titchtech@titchtechacademyserver:~$ sudo systemctl status wazuh-indexer.service
● wazuh-indexer.service - Wazuh-indexer
     Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-04-19 03:53:00 UTC; 8min ago
       Docs: https://documentation.wazuh.com
   Main PID: 3751 (java)
      Tasks: 87 (limit: 9346)
     Memory: 4.3G (peak: 4.3G)
        CPU: 1min 26.550s
     CGroup: /system.slice/wazuh-indexer.service
             └─3751 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=10 -XX:+Al

Apr 19 03:52:42 titchtechacademyserver systemd[1]: Starting wazuh-indexer.service - Wazuh-indexer...
Apr 19 03:52:45 titchtechacademyserver systemd-entrypoint[3751]: WARNING: A terminally deprecated method in java.lang.System has been called
Apr 19 03:52:45 titchtechacademyserver systemd-entrypoint[3751]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.OpenSearch (file:/usr/sh
Apr 19 03:52:45 titchtechacademyserver systemd-entrypoint[3751]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.OpenSearch
Apr 19 03:52:45 titchtechacademyserver systemd-entrypoint[3751]: WARNING: System::setSecurityManager will be removed in a future release
Apr 19 03:52:46 titchtechacademyserver systemd-entrypoint[3751]: WARNING: A terminally deprecated method in java.lang.System has been called
Apr 19 03:52:46 titchtechacademyserver systemd-entrypoint[3751]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.Security (file:/usr/shar
Apr 19 03:52:46 titchtechacademyserver systemd-entrypoint[3751]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.Security
Apr 19 03:52:46 titchtechacademyserver systemd-entrypoint[3751]: WARNING: System::setSecurityManager will be removed in a future release
Apr 19 03:53:00 titchtechacademyserver systemd[1]: Started wazuh-indexer.service - Wazuh-indexer.
lines 1-21/21 (END)
```

Figure 8. Check Indexer Service Status

6.3 Agent Installation

- Installed Wazuh agents on monitored systems
- Connected agents to Wazuh Manager

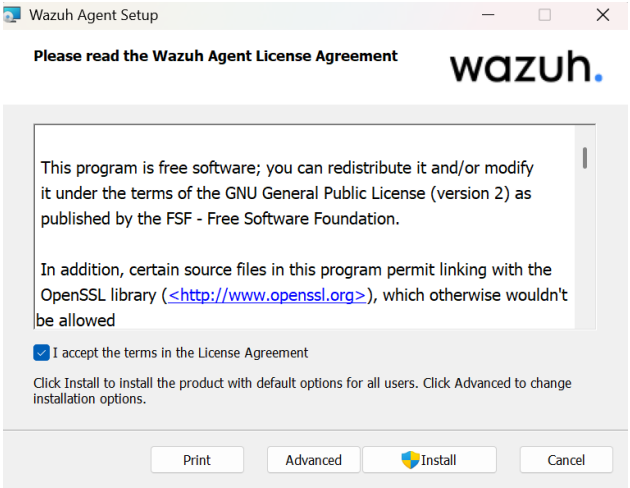

Figure 12.Initial Agent Installation GUI

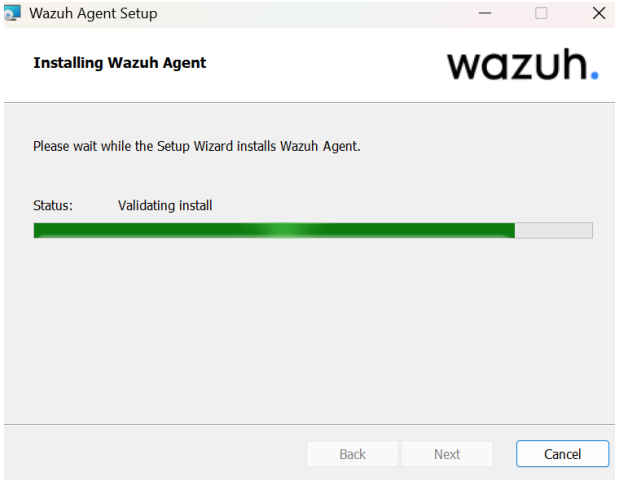

Figure 11. Agent Installation In Progress

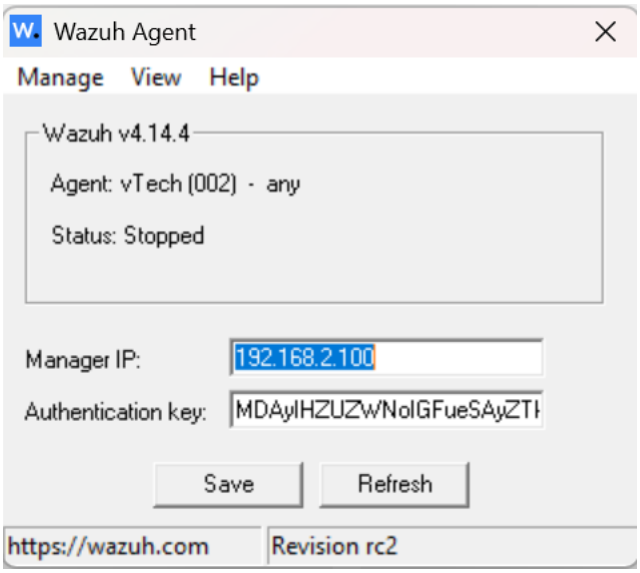

Figure 13.Configuring Agent with the Wazuh Server IP and Agent Keys

6.4 Agent Connectivity Verification

- Verified agents are actively sending logs

```
root@titchtechacademyserver:/home/titchtech# /var/ossec/bin/agent_control -l 001

Wazuh agent_control. List of available agents:
   ID: 000, Name: titchtechacademyserver (server), IP: 127.0.0.1, Active/Local
   ID: 001, Name: windows11_titch, IP: 192.168.2.101, Active

List of agentless devices:

root@titchtechacademyserver:/home/titchtech#
```

Figure 14. Verify Agent Activeness

[Insert Screenshot: Agent Status (Active/Connected) Here]

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7. Log Collection & Monitoring

The system collects logs from:

- System logs (Linux/Windows)
- Authentication logs
- Network activity
- Security events

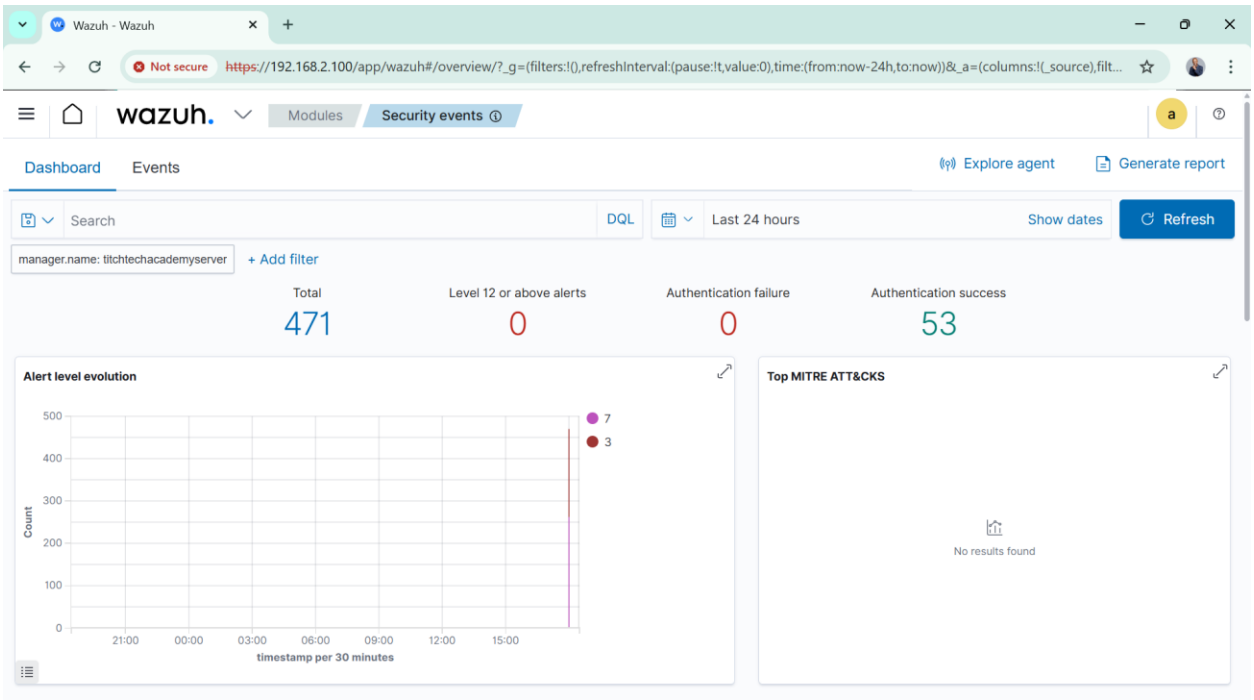

Figure 15. Security log Overview

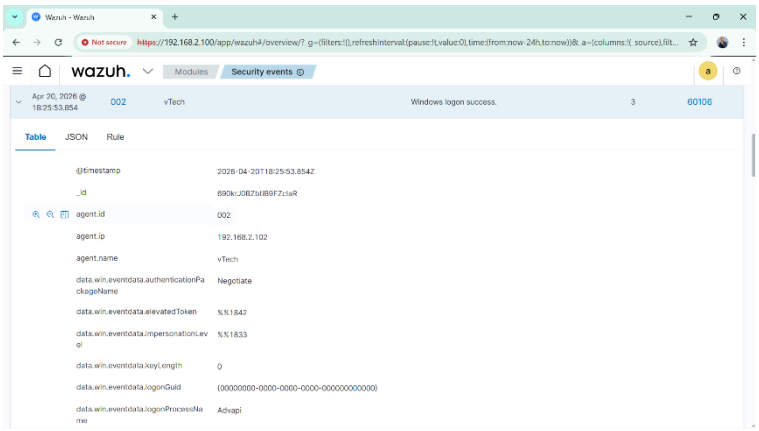

Figure 16. Security logs Collected

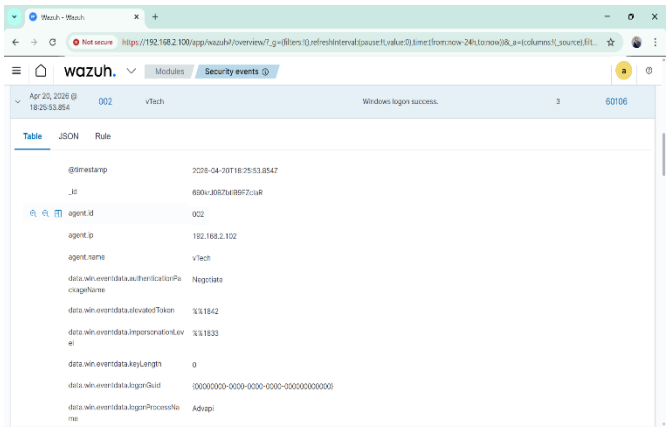

Figure 17. Detailed information about a logon event

8. Detection & Alerting

Wazuh analyzes logs using predefined rules to detect:

- Brute force attacks
- Unauthorized access attempts
- Malware indicators
- Suspicious system activity

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9. Testing & Validation

Test Scenarios Performed

- Failed login attempts (Brute force simulation)
- Unauthorized file access
- Suspicious command execution

Results

- Alerts were generated successfully
- Logs were collected in real time
- System responded as expected

---

10. Challenges Encountered

- Working in a TTY-only environment required strong command-line skills
- Network configuration issues during setup
- Agent connectivity troubleshooting

---

11. Solutions Implemented

- Used SSH for remote management
- Verified firewall rules and open ports
- Ensured correct agent configuration (IP and keys)

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12. Security Considerations

- Dedicated server reduces attack surface
- TTY-only Ubuntu minimizes unnecessary services
- Secure communication between agents and server

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13. Key Achievements

- Successfully deployed a centralized SIEM system
- Achieved real-time log monitoring and alerting
- Improved visibility into system activities
- Built a scalable security monitoring foundation

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14. Conclusion

The implementation of Wazuh on a dedicated Ubuntu Server (TTY-only) system provides a robust and efficient centralized security monitoring solution. It enhances the organization's capability to detect and respond to cyber threats in real time.

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15. Future Improvements

- Integrate Elastic Stack for advanced visualization
- Deploy more agents across the network
- Implement automated incident response
- Add threat intelligence feeds

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16. References

- Wazuh Documentation
- Ubuntu Server Documentation

---

17. Appendix

Commands Used

```
# Update system
sudo apt update && sudo apt upgrade -y

# Install Wazuh
curl -sO https://packages.wazuh.com/4.x/wazuh-install.sh
sudo bash wazuh-install.sh

# Check status
sudo systemctl status wazuh-manager
```